Xin Liu

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My research focuses on wave optics modeling, light shaping, and joint optimization of optical systems and reconstruction algorithms for imaging and sensing.

Computational imaging · Differentiable optics · Inverse problems · Machine learning · Diffraction · Microscopy

Academic Appointments

04/2024 – Present **Postdoctoral Researcher** Hong Kong SAR, China
Department of Electrical and Computer Engineering, The University of Hong Kong (HKU)

01/2024 – 03/2024 **Research Assistant** Hangzhou, China
State Key Laboratory of Extreme Photonics and Instrumentation, Zhejiang University (ZJU)

Education

09/2018 – 12/2023 **Ph.D.** in Optical Engineering Hangzhou, China
State Key Laboratory of Extreme Photonics and Instrumentation, Zhejiang University (ZJU)
Dissertation: *Research on the principles and key issues of point spread function engineering*

09/2014 – 06/2018 **B.E.** in Optoelectronic Information Science and Engineering Guangzhou, China
School of Physics and Optoelectronics, South China University of Technology (SCUT)

Selected Publications

[#] equal contribution; [*] corresponding author.

[11] Retained accuracy with reduced precision in wave propagation modeling.
Xin Liu and Yifan Peng.
Photonics Research, 14(5), 1762–1774 (2026).

[10] Learned off-aperture encoding for wide field-of-view RGBD imaging.
Haoyu Wei, **Xin Liu**, Yuhui Liu, Qiang Fu, Wolfgang Heidrich, Edmund Y. Lam, and Yifan Peng.
IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 1–13 (2025).

[9] In situ fully vectorial tomography and pupil function retrieval of tightly focused fields.
Xin Liu, Shijie Tu, Yiwen Hu, Yifan Peng, Yubing Han, Cuifang Kuang, Xu Liu, and Xiang Hao.
Nature Communications, 16(1), 3478 (2025).

[8] Diffraction modeling between arbitrary non-parallel planes using angular spectrum rearrangement.
Yiwen Hu[#], **Xin Liu[#*]**, Xu Liu, and Xiang Hao[*].
Optica, 12(1), 39–45 (2025). [**Monthly top downloads, January**]

[7] Modeling off-axis diffraction with the least-sampling angular spectrum method.
Haoyu Wei[#], **Xin Liu[#]**, Xiang Hao, Edmund Y. Lam, and Yifan Peng.
Optica, 10(7), 959–962 (2023). [**Monthly top downloads, July & August**]

[6] Fast generation of arbitrary optical focus array.
Xin Liu, Yiwen Hu, Shijie Tu, Cuifang Kuang, Xu Liu, and Xiang Hao.
Optics and Lasers in Engineering, 162, 107405 (2023).

[5] Investigating deep optics model representation in affecting resolved all-in-focus image quality and depth estimation fidelity.
Xin Liu[#], Linpei Li[#], Xu Liu, Xiang Hao, and Yifan Peng.
Optics Express, 30(20), 36973–36984 (2022).

Selected Publications (continued)

[4] Calibration of phase-only liquid-crystal spatial light modulators by diffractogram analysis.
Xin Liu, Shijie Tu, Cuifang Kuang, Xu Liu, and Xiang Hao.
Optics and Lasers in Engineering, 156, 107056 (2022).

[3] Accurate background reduction in adaptive optical three-dimensional stimulated emission depletion nanoscopy by dynamic phase switching.
Shijie Tu, **Xin Liu**, Difu Yuan, Wenli Tao, Yubing Han, Yan Shi, Yanghui Li, Cuifang Kuang, Xu Liu, Yufeng Yao, Yesheng Xu, and Xiang Hao.
ACS Photonics, 9(12), 3863–3868 (2022). [**Cover article**]

[2] Generation of arbitrary longitudinal polarization vortices by pupil function manipulation.
Xin Liu, Yifan Peng, Shijie Tu, Jun Guan, Cuifang Kuang, Xu Liu, and Xiang Hao.
Advanced Photonics Research, 2(1), 2000087 (2021). [**Cover article**]

[1] Aberrations in structured illumination microscopy: A theoretical analysis.
Xin Liu, Shijie Tu, Yan Xu, Hongya Song, Wenjie Liu, Qiulan Liu, Cuifang Kuang, Xu Liu, and Xiang Hao.
Frontiers in Physics, 7, 254 (2020).

Awards and Honors

2026	Outstanding Achievements Award The 6th International Conference on Computational Imaging (CITA 2026) Chinese Society for Optical Engineering
2025	Outstanding Young Rising Star Award (Ranked 1st among 20 presenters) The 5th International Conference on Computational Imaging (CITA 2025) Chinese Society for Optical Engineering
2024	Honorable Mention The 4th International Conference on Computational Imaging (CITA 2024) Chinese Society for Optical Engineering
2022	Scholarship for Innovation and Entrepreneurship, Zhejiang University
2021, 2022	Merit Graduate Student, Zhejiang University
2020 – 2022	Junshi Chen Scholarship, Zhejiang University
2020, 2021	Outstanding Graduate Student Cadre, Zhejiang University
2019, 2021, 2022	Outstanding Graduate Student, Zhejiang University

Invited Talks

09/2026	Scaling differentiable optical simulation The 6th International Conference on Computational Imaging (CITA 2026)	Shenzhen, China
09/2025	Exploring advanced imaging with computational optics The 5th International Conference on Computational Imaging (CITA 2025)	Suzhou, China
09/2024	Free-space wave propagation modeling with the least-sampling angular spectrum method The 4th International Conference on Computational Imaging (CITA 2024)	Xiamen, China
05/2024	Exploring wave propagation modeling in free space Guest lecture, King Abdullah University of Science and Technology	Thuwal, Saudi Arabia

Research Mentoring

2021 – 2026	Multi-dimensional quantitative characterization methods for metasurface-modulated optical fields Xinjie Sun Zhejiang University	Ph.D. thesis
2022 – 2025	Learned wide-angle imaging with optical modeling and optimization Haoyu Wei The University of Hong Kong	Ph.D. thesis

Research Mentoring (continued)

2021 – 2025	Characterization of optical fields with computational methods Yiwen Hu Zhejiang University	Ph.D. thesis
2023	Deep stereo optics imaging for high-fidelity depth estimation Yuhui Liu The University of Hong Kong	Master's thesis
2022	Full vectorial manipulation of tightly focused optical fields Haiwei Wang Zhejiang University	Bachelor's thesis
2022	Numerical simulation of vectorial fields based on vectorial diffraction theory Yunpeng Wang Changchun University of Science and Technology	Bachelor's thesis

Academic Service

Guest Editor	Computational Methods for Advanced Optical Imaging, Photonics, MDPI, 2026
Area Chair	International Conference on Computational Photography (ICCP), 2026
Program Committee	The 5th International Conference on Computational Imaging (CITA 2025)
Program Chair	HKU-KAUST Joint Postgraduate Workshop on Computational Imaging, 2025
Journal Reviewer	Optica; Advanced Functional Materials; Light: Advanced Manufacturing; Laser & Photonics Reviews; Advanced Science; Optics Letters; Optics Express; JOSA A; JOSA B; Applied Optics; Journal of Microscopy; Optics Communications
Conference Reviewer	SIGGRAPH; SIGGRAPH Asia; ICCP

Patents

2026	Imaging system for capturing wide field-of-view images Haoyu Wei, Yifan Peng, Yin Mun Edmund Lam, and **Xu Liu** Granted patent: US 2026/0339114 A1
2022	A method and device for calibration of phase-only spatial light modulators Xiang Hao, **Xin Liu**, Yubing Han, Cuifang Kuang, and Xu Liu Granted patent: CN112697401B
2020	A method for point spread function reconstruction Xiang Hao, **Xin Liu**, Cuifang Kuang, and Xu Liu Granted patent: CN110133849B
2020	A two-dimensional rapid beam scanning module for confocal fluorescence microscopy Xiang Hao, **Xin Liu**, Shijie Tu, and Xu Liu Granted patent: CN109491065B
2020	A device for common-path beam modulation in imaging and lithography systems Xu Liu, **Xin Liu**, Xiang Hao, Cuifang Kuang, and Haifeng Li Granted patent: CN110568650B
2020	A device for generating super-resolution optical focus arrays Xiang Hao, **Xin Liu**, Cuifang Kuang, Xu Liu, and Haifeng Li Granted patent: CN110568731B
2020	A multicolor super-resolution microscope with automatic alignment capability Xiang Hao, Shijie Tu, **Xin Liu**, and Xu Liu Granted patent: CN109358030B